ELEKTRA

SMALL FOOT PRINT SERIES

Lithium-Iron (LiFePo4) UPS

I.T GRADE ONLINE DOUBLE CONVERSION UPS

LITHIUM - ION (LiFePo4) COMPATIBLE

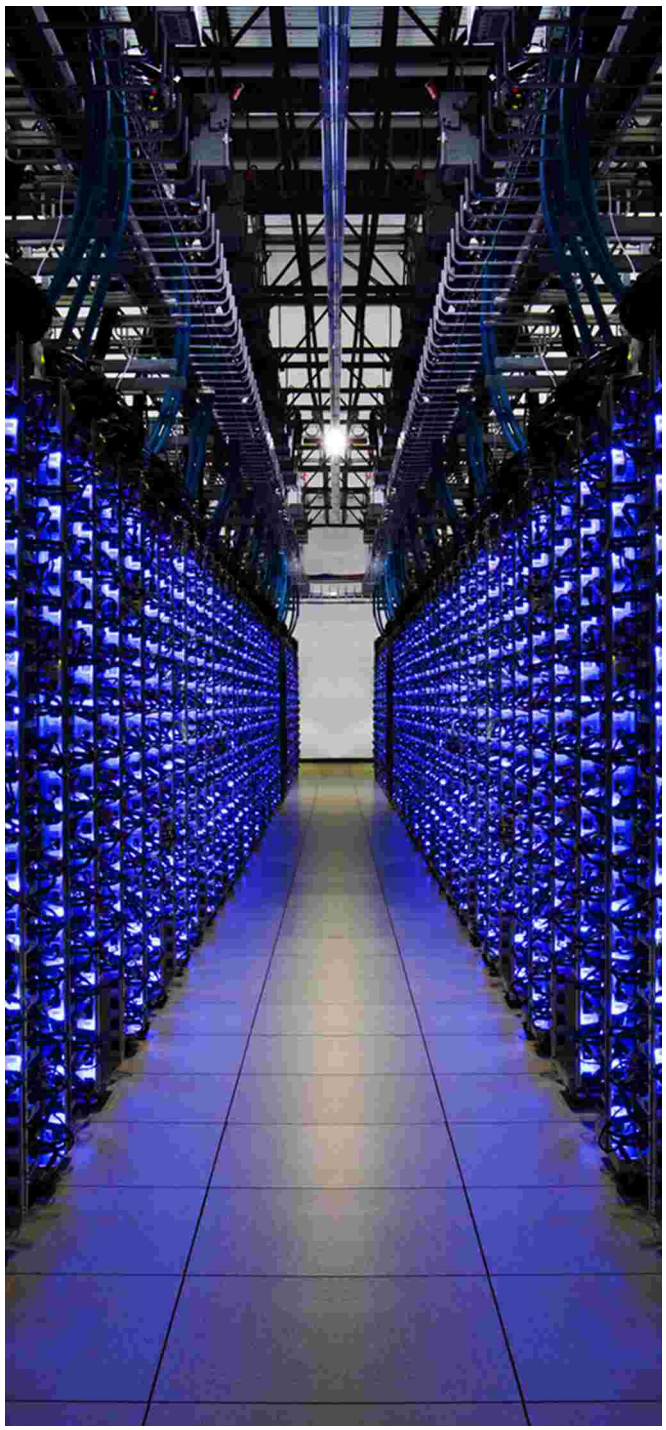
10KVA ~ 200KVA (3/3)

UNITY POWER FACTOR

Mission Critical &
I.T Grade UPS

I.T GRADE ONLINE DOUBLE CONVERSION UPS (SMALL FOOT PRINT)

DP Electronics (Deutsche Power Co., Limited)



Elektra Series (3/3, H.F) (10KVA~200KVA)

Elektra Series I.T Grade On Line Double Conversion UPS represents the best solution that is both effective and environment friendly, it delivers and excellent integrated autonomy, reduced footprint and optimum input and output performances, guaranteeing reliable and flexible power.

Elektra Double conversion pure on-line three phase input and output UPS adopt double conversion technology, able to provide solutions to conditions like voltage instantaneous drop-off or damping concussion, high-voltage plus, voltage pulsation, surge voltage, harmonic distortion, and frequency fluctuation, guarantees a safe and reliable power.

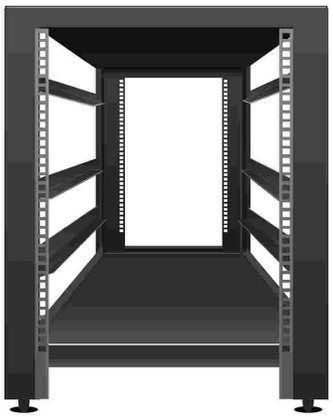
UPS status can be monitored at a glance on an intuitive LCD screen.

Elektra Series offer redundant and capacity parallel Ups, the right solution for all applications requiring a perfect and uninterrupted power supply.

This is a green product that comply with the products pollution control management measures, the product under normal use, will not harm the environment and personals using it.



I.T GRADE ONLINE DOUBLE CONVERSION UPS (SMALL FOOT PRINT)





Power Factor 1

High adoptability for load as it can connect full inductive or capacitive load.
Deutsche Power ESHS Series has output power factor 1 and UPS supply power to 100% Unbalance load

High-efficient, Energy Saving, Green Environmental Protection

The efficiency of the machine is as high as 94% and reaches over 98% in the ECO mode which can help to reduce the power loss of the UPS's IGBT, FM, intelligent rectifier and inverter technology, input power factor 0.99, & input current harmonic wave (THDi) is as low as 2%.

Excellent Overload Capacity

When higher protection required, the optional dust-proof accessories can help to increase the protection level in harsh environment and safeguard the UPS's security. Elektra series UPS provide a broader range of input voltage from 208VAC to 478VAC, which ensures to be applicable in the unpleasant grid environment of different power supply equipment like generator, with its extraordinary input frequency.

Zero Switching

Transfer time when UPS transferred to the battery or vice versa is zero, effectively guarantee the load operation security and reliability.

Application

ESHS Series range provide a combination of outstanding protection features and flexibility, making it the right choice for applications demanding optimum reliability and energy efficiency.

Systems such as servers, networking devices, workstations, storage systems and various IT equipment find the right protection element in Elektra Series, especially, when combined with Elektra Powerful Connectivity suite.

VolP equipment, railway control systems, medical laboratory instrumentation, and many other industrial applications may also benefit from the consistent and high quality power provided by the Elektra, thanks to the robustness, precision and high efficiency provided.

Key Features

- Online-Double conversion
- Output transfer time is Oms
- PFC technology
- Full digital control (DSP)
- Output power factor: 1
- Input current harmonic: 3%
- ECO function
- Charging/Rectifier/Inverter fully digital control technology
- Optimization battery group, the quantity of battery: 10- 30K (16/18/20/32/34/36/38/40 pes optional) 40-200K (32/34/36/38/40 pes optional)
- Wide input voltage range: 208-478Vac
- Wide input frequency range: 40-70Hz
- DC start
- Communication port: USB / RS232 / RS485 /
- Parallel port / Dry contact
- Options: SNMP card/Relay card
- LCD I LED double display
- Intelligent charging management
- EPO function
- Common battery group
- The output can meet 100% unbalanced load

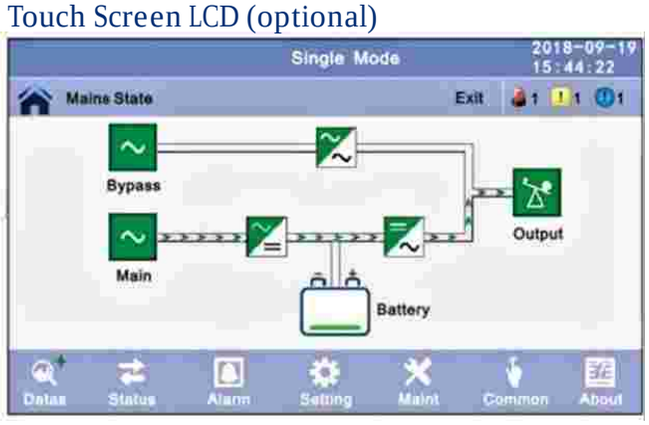
Powerful Extensibility Features

Smart slot provides rich scalable features, USB can be selected, AS400 card, SNMP card, RS485 card and environmental monitoring card.

Standards

Safety:
IEC/EN62040-1, IEC/EN60950-1. EMC IEC/ EN62040-2, IEC61000-4-2, IEC61000-4-3, IEC61000-4-4, IEC61000-4-5, IEC61000-4-6, IEC61000-4-8
Noise Suppression
Complies with EN62040~2

Touch Screen LCD (optional)



DP Electronics (Deutsche Power Co., Limited)

I.T GRADE ONLINE DOUBLE CONVERSION UPS (SMALL FOOT PRINT)

DP Electronics (Deutsche Power Co., Limited)

TECHNICAL SPECIFICATION FOR THREE PHASE IN & THREE PHASE OUT

MODEL	ESHS1033	ESHS1533	ESHS 2033	ESHS3033	ESHS4033
Rating	10KVA	15KVA	20KVA	30KVA	40KVA
INPUT					
Nominal voltage	380/400/415Vac, (3Ph+N+PE)				
Operating voltage range	138~485Vac for 40%load; 305~485Vac for full-load				
Operating frequency range	40~70Hz				
Power factor	≥1				
Bypass voltage range	380Vac Max. voltage: +25%(optional +1 0%,+15%,+20%) 400Vac Max. voltage: +20%(optional +1 0%,+15%) 415Vac Max.voltage: +15%(optional +10%) Min. voltage: -45% (optional -20%,-30%)				
Bypass frequency range	Frequency synchronize tracing range: ± 10%				
ECO range	Same as bypass				
Harmonic distortion (THDi)	≤3% (100%non-linear load)				
OUTPUT					
Output voltage	380/400/415Vac (3Ph+N+PE)				
Power factor	1				
Voltage regulation	±1%				
Frequency: line mode bat.mode	± 1%/ ± 2%/ ± 4%/ ± 5%/ ± 10% of the rated frequency(optional) 50/60(± 0.1)Hz				
Crest factor	3:1				
Harmonic distortion (THD)	2% with linear load 5% with non linear load				
Efficiency	94.5%				
BATTERY	SLA / VRLA / LiFePo4				
Battery Voltage	±120Vdc(20pcs 12V9AH); Long run unit: ±96V/±108V/ ±120Vdc(16/18/ 20pcs optional)	Standard unit:±120Vdc (2x20pcs 12V9AH); Long run unit Optional Voltage:±96VI±108VI± 120Vdc(16/18/20pcs optional)	±120Vdc (3x20pcs12V9AH) Long run unit Optional Voltage:±96V/± 108V/±120Vdc (16/18/20pcs optional)	Optional Voltage:±192VI± 204VI±216VI±228VI ±240Vdc(32/34/36/38/40pcs optional)	
Charge Current(A) (charge current can be set according to battery capacity installed)	Standard unit: 1.35A Long run unit: Max.current 1OA	Standard unit: 2.7A Long run unit: Max.current 1OA	Standard unit: 4.05A Long run unit: Max.current 15A	Max.current 15A	
SYSTEM FEATURES					
Transfer time	Utility to Battery: Oms; Utility to bypass: Oms				
Over Load	Load:≤; 110%: last 60min,:≤; 125%: last 1Omin,:≤; 150%: last 1min, ≥150% change to bypass.				
Short Circuit	Hold Whole Syetem				
Communication	USB,RS232, RS485, Parallel port, REPO port, Coupler dry contact, Intelligent slot, SNMP card				
	(optional), Relay card (optionai).LBS port (only 60-80k)				
ENVIRONMENTAL					
Operating temperature	0-40"C				
Storage temperature Humidity range	-25- 55"C(No battery) 0-95% (Non condensing)				
Altitude	< 1500m.When>1500m,lower the rated power for use				
Noise level	<55dB	<58dB			
PHYSICAL					
Dimension D x W x H (mm)	828x250x868				
Net Weight (kg)	115/57	170/63	171/64	223/71	
PHYSICAL					
Saftey	IEC/EN62040-1 ,IEC/EN60950-1				
EMC	IEC/EN62040-2,1EC61 000-4-2,1EC61 000-4-3,1EC61 000-4-4,1EC61 000-4-5,1EC61 000-4-6,1EC61 000-4-8				

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I.T GRADE ONLINE DOUBLE CONVERSION UPS (SMALL FOOT PRINT)

DP Electronics (Deutsche Power Co., Limited)

TECHNICAL SPECIFICATION FOR THREE PHASE IN & THREE PHASE OUT

MODEL	ESHS5033	ESHS6033	ESHS10033	ESHS12033	ESHS16033	ESHS18033
Rating	50KVA	60KVA	100KVA	ESHS12033	160KVA	180KVA
INPUT						
Nominal voltage	380/400/415Vac, (3Ph+N+PE)					
Operating voltage range	138~485Vac for 40%load; 305~485Vac for full-load					
Operating frequency range	40~70Hz					
Power factor	≥1					
Bypass voltage range	220Vac Max.voltage: +25%(optional +10%,+15%,+20%) 230Vac Max.voltage: +20%(optional +10%,+15%) 240Vac Max.voltage: +15%(optional +10%) Min. voltage: -45% (optional -20%,-30%) Frequency synchronize tracing range: ±10%					
Generator input	Support					
Harmonic distortion (THDi)	≤3% (100% linear load)					
OUTPUT						
Output voltage	380/400/415Vac (3Ph+N+PE)					
Power factor	1					
Voltage regulation	±1%					
Frequency: line mode bat.mode	Synchronize with input; when input frequency >±10% (±1%/±2%/±4%/±5% optional), output (50/60±0.1)Hz. (50/60±0.1)Hz					
Crest factor	3:1					
Harmonic distortion (THD)	1% with linear load, 4% with non linear load					
Efficiency	96%					
BATTERY	SLA / VRLA / LiFePo4					
Battery Voltage	Optional Voltage: ±180V/±192V/±204V/±216V/±228V/±240/±252/±264/±276/±288/±300Vdc(30/32/34/36/38/40/42/44/46/48/50pcs optional); 360Vdc~600Vdc (30~50 pcs, 36 pcs default, 36 to 50 pcs output power factor1.0; 32~34 pcs output power factor 0.9					
Charge Current(A)	Max.current 20A		Max.current 40A		Max.current 60A	
SYSTEM FEATURES						
Transfer time	Utility to Battery : 0ms; Utility to bypass: 0ms					
Over Load INV. Mode Bypass Mode	110% overload for 60 min; 125% overload for 10 min; 150% overload for 1 min 135% overload for long time; >1000% overload for 100 ms					
Backfeed	Support					
Alarm	Overload, utility abnormal, UPS fault, battery low, etc.					
Protection	Short circuit, overload, over temperature, battery low, fan fault alarm.					
Remote LCD	Support					
Communication	USB, RS485, Parallel port, Coupler dry contact, Intelligent slot, SNMP card (optional), Relay card (optional)					
ENVIRONMENTAL						
Operating temperature	0 C ~ 40 C					
Storage temperature	25 C~55 C (no battery)					
Humidity range	0~95% (noncondensing)					
Altitude	< 1500m.When>1500m,lower the rated power for use					
Noise level	<55dB	<58dB	<60dB		<63dB	<65dB
ENVIRONMENTAL						
Dimension D x W x H (mm)	828x250x868		850x450x1200			
Net Weight (kg)	68	72	115	120		
PHYSICAL						
Saftey	IEC/EN62040-1 ,IEC/EN60950-1					
EMC	IEC/EN62040-2,1EC61 000-4-2,1EC61 000-4-3,1EC61 000-4-4,1EC61 000-4-5,1EC61 000-4-6,1EC61 000-4-8					

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I.T GRADE ONLINE DOUBLE CONVERSION UPS (SMALL FOOT PRINT)

DP Electronics (Deutsche Power Co., Limited)

TECHNICAL SPECIFICATION FOR SINGLE PHASE / THREE PHASE IN & SINGLE PHASE / THREE PHASE OUT

MODEL	ESHS1033RT*	ESHS1533RT*	ESHS2033RT*	ESHS2533RT	ESHS3033RT	ESHS4033RT	ESHS5033RT
Rating	10KVA	15KVA	20KVA	25KVA	30KVA	40KVA	50KVA
INPUT							
Nominal voltage	220/230/240Vac (L+N+PE)* & 380/400/415Vac, (3Ph+N+PE)						
Operating voltage range	208~478Vac						
Operating frequency range	40Hz~70Hz						
Power factor	0.99 & 1						
Bypass voltage range	220Vac Max.voltage: +25%(optional +10%,+15%,+20%) 230Vac Max.voltage: +20%(optional +10%,+15%) 240Vac Max.voltage: +15%(optional +10%) Min. voltage: -45% (optional -20%,-30%) Frequency synchronize tracing range: ±10%						
Generator input	Support						
Harmonic distortion (THDi)	<3% (100% linear load)						
OUTPUT							
Output voltage	220/230/240Vac (L+N+PE)* & 380/400/415Vac (3Ph+N+PE)						
Power factor	1						
Voltage regulation	±1%						
Frequency: line mode bat.mode	Synchronize with input; when input frequency >±10% (±1%/±2%/±4%/±5% optional), (50/60±0.1)Hz.						
Crest factor	3:1						
Harmonic distortion (THD)	2% with linear load, 4% with non linear load						
Efficiency	95.60%						
BATTERY	SLA / VRLA / LiFePo4						
Battery Voltage	Optional Voltage: ±180V/±192V/±204V/±216V/±228V/±240/±252/±264/±276/±288/±300Vdc (30/32/34/36/38/40/42/44/46/48/50pcs optional); 360Vdc~600Vdc (30~50 pcs, 30 pcs define, 36~50 pcs no power derating; 32~34 pcs output power factor 0.9; 30 pcs output power factor 0.8;)						
Charge Current(A)	Max.current 18A					Max.current 20A	
SYSTEM FEATURES							
Transfer time	Utility to Battery : 0ms; Utility to bypass: 0ms						
Over Load	Load<110% last 60 min; <125% Last 10 min; 150% Last 1 min						
Backfeed	Support						
Alarm	Overload, utility abnormal, UPS fault, battery low, etc.						
Protection	Short circuit, overload, over temperature, battery low, fan fault alarm.						
Communication 10-30Kva	USB, RS232, RS485, Parallel port, Dry contact port, REPO port, Backfeed port, Intelligent slot, SNMP (optional)						
40-50Kva	USB, RS232, RS485, Parallel port, REPO port, LBS port, Backfeed port, Intelligent slot, SNMP card (optional), Relay card (optional)						
ENVIRONMENTAL							
Operating temperature	0 C ~ 40 C						
Storage temperature	25 C~55 C (no battery)						
Humidity range	0~95% (non condensing)						
Altitude	< 1500m.When>1500m,lower the rated power for use						
Noise level	<55dB					<56dB	<58dB
PHYSICAL							
Dimension D x W x H (mm)	443x580x131(3U)					800x440x175(4U)	
Net Weight (kg)	25	27-28				45	48
STANDARDS							
Saftey	IEC/EN62040-1 ,IEC/EN60950-1						
EMC	IEC/EN62040-2,1EC61 000-4-2,1EC61 000-4-3,1EC61 000-4-4,1EC61 000-4-5,1EC61 000-4-6,1EC61 000-4-8						

*81/1,3/1,3/3 Can be achieved in 10,12,20 Kva Ups by LCD interface

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DP Electronics (Deutsche Power Co., Limited)

LiFePo4 Battery Pack

Deutsche Power provides safe LiFePO4 battery solutions for UPS and Energy storage system. This ba ttery system serves for UPS and Energy storage system with perf ect compatibility and sa fe long cy cle life.





Features:

- Longer Cycle Life
- Lighter Weight
- Higher Power
- Wider Temperature Range
- Superior Safety

LIFEPO4 BATTERY SPECIFICATION

Common Models	348VDC			480VDC			N Series
Models	CLFP380-20-H	CLFP380-50-H	CLFP380-100-H	CLFP480-20-H	CLFP480-50-H	CLFP480-100-H	CLFPXXX-YY-H
Nominal Voltage	384VDC			480VDC			384~480*N
Capacity AH	20 AH	50 AH	100 AH	20 AH	50 AH	100 AH	20AH~100*N
Energy	7.6KWH	19 KWH	38.4 KWH	9.6 KWH	24 KWH	48KWH	7.6KWH~100KWH*N
Cluster	1	2	3	1	2	3	1~3*N
Adjustable Voltage Range	380-448 V			480~560 V			380~560V

Common Models	CLFP537-100-4	CLFP537-200-4	CLFP537-300-4	CLFP537-100-4*N-H
Nominal Voltage	537.6V			
Cluster	1	2	3	N
Adjustable Rate Voltage	500~800V			
Nominal Capacity	100Ah	200Ah	300Ah	100Ah~300Ah*N
Energy	53.76Kwh	107.52Kwh	161.28Kwh	53.76~161.28Kwh*N
Cycle Life	5000 times@25C 1C/1C 80%DOD			
Charge Voltage	596V			
Adjustable Charge Voltage	500V~900V			

For detail please check HV catalogue

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Battery Pack

Deutsche Power Elektra series battery pack , provides extended runtime for mission critical applications. All Battery Pack are 384VDC and can be slecetd from 2 Models as per eequired backup time .The Number of Bank can be connected to N+X quanity and as per required autonomy . It is a lead-acid battery with a 2-year warranty. It is a lead-acid battery . These Battery Packs are 19" and can fit in any Rack with 3U Size .



DP Electronics (Deutsche Power Co., Limited)

BATTERY PACK SPECIFICATION

Model	RT-BR3207	RT-BR2007	RT-BR3212
Product or Component Type	Replacement battery cartridge	Replacement battery cartridge	Replacement battery cartridge
Battery Type	Lead-Acid battery	Lead-Acid battery	Lead-Acid battery
Provided Equipment	Battery Pack	Battery Pack	Battery Pack
Battery Modules	2	1	3
Battery Type	12Vx07AH	12Vx07AH	12Vx12AH
autonomy	Depends on Load	Depends on Load	Depends on Load
Device mounting	Rack & Floor	Rack & Floor	Rack & Floor
Battery Voltage	384VDC	240VDC	384VDC
Color	Black	Black	Black
Height	5.2"	5.2"	5.2"
Width	19"	19"	19"
Depth	22.2"	22.2"	22.2"
Net Weight	15KGS	12KGS	15KGS
Mounting Mode	Rack & Floor	Rack & Floor	Rack & Floor
Ambient Air Temperature for Operation	32...104 °F (0...40 °C)	32...104 °F (0...40 °C)	32...104 °F (0...40 °C)
Operating altitude	0...10000 ft.	0...10000 ft.	0...10000 ft.
Relative Humidity	0...95 % non-condensing	0...95 % non-condensing	0...95 % non-condensing
Ambient Air Temperature for Operation	5...113 °F (-15...45 °C)	5...113 °F (-15...45 °C)	5...113 °F (-15...45 °C)
Storage altitude	0...50000 ft. (0...15240 m)	0...50000 ft. (0...15240 m)	0...50000 ft. (0...15240 m)
Storage Relative Humidity	0...95 % non-condensing	0...95 % non-condensing	0...95 % non-condensing

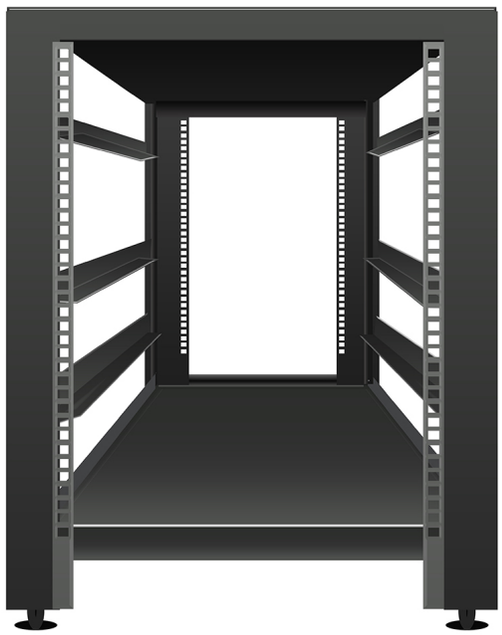
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Battery Rack

Deutsche Power rack uninterruptible power supply (UPS) system is designed for use within standard 19-inch IT racks or rack-based enclosures. With 15U Size

Racks can be used in a wide variety of locations and environments meaning that a rackmount UPS must be able to function properly in these locations as well. They can be located anywhere from large data centers to network closets and other remote edge locations.

Rackmount models can be mounted in standard 19? rack enclosures and can require anywhere from 1U to 12U (rack space).



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BATTERY RACK SPECIFICATION

Product or Component Type	UPS Rack
TYPE	Rack Version
Provided Equipment	UPS and Battery Rack
UPS / Battery Modules in 1 Rack	5
Device mounting	Tower
Color	Black
Height	31"
Width	23"
Depth	32"
Net Weight	50KG
Mounting Mode	Tower

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